REPUBLIC OF CAMEROON — Peace · Work · Fatherland

« RÉUSSIR SON EXAMEN » APP — GCE SECTION

GCE O LEVEL — MATHEMATICS

Practice Paper WITH FULL SOLUTIONS — 1 h 30

SECTION A : QUESTIONS

1. Evaluate $2^3 \times 3^2$.
2. Solve for x : $3x + 5 = 20$.
3. Factorise completely : $x^2 - 9$.
4. Find the simple interest on 50 000 FRS for 3 years at 5 % per annum.
5. A circle has radius 14 cm (take $\pi = 22/7$). Find (a) its circumference, (b) its area.
6. Find the mean of the numbers : 4 ; 7 ; 9 ; 10 ; 15.
7. A right-angled triangle has legs 9 cm and 12 cm. Find the hypotenuse.
8. A bag contains 3 red balls and 5 blue balls. A ball is drawn at random. Find the probability that it is red.

SECTION B : FULL SOLUTIONS

1. $2^3 = 8$ and $3^2 = 9$, so $2^3 \times 3^2 = 8 \times 9 = 72$.
2. $3x + 5 = 20 \rightarrow 3x = 20 - 5 = 15 \rightarrow x = 15 \div 3 = 5$.
3. $x^2 - 9$ is a difference of two squares : $x^2 - 3^2 = (x - 3)(x + 3)$.
4. $I = (P \times R \times T) / 100 = (50\,000 \times 5 \times 3) / 100 = 7\,500$ FRS.
5. (a) $C = 2\pi r = 2 \times 22/7 \times 14 = 88$ cm. (b) $A = \pi r^2 = 22/7 \times 14 \times 14 = 616$ cm².
6. Sum = 4 + 7 + 9 + 10 + 15 = 45. Mean = 45 ÷ 5 = 9.
7. $h^2 = 9^2 + 12^2 = 81 + 144 = 225$, so $h = \sqrt{225} = 15$ cm (Pythagoras' theorem).
8. Total balls = 3 + 5 = 8. P(red) = number of red / total = 3/8.